Date: 22^{nd} April 2026 DAB_II Section:A

Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA) is a statistical technique used to compare the means of three or more groups simultaneously to determine whether there is a significant difference among them. Instead of performing multiple pairwise comparisons, ANOVA evaluates the overall variation in the data by separating it into variation within groups and variation between groups. By comparing these sources of variation, ANOVA helps identify whether at least one group mean differs significantly from the others, while controlling the overall probability of making incorrect conclusions.

A, B, C➔ t test➔ (A,B), (A,C), (B,C)➔3 t-test➔ 0.05*3= 0.15➔15% ➔ t test worth it

A:20➔5

B:21➔10

C:15➔25

Why significance level increases with multiple t-tests (example: 4 means)

If you have **4 (A,B,C & D) groups**, the number of pairwise t-tests is:

$$\frac{n(n-1)}{2} = \frac{4(3)}{2} = 6$$

So you would perform **6 separate t-tests**.

(A,B),(A,C), (A,D), (B,C),(B,D), (C,D)

Each test might use a significance level of **α = 0.05**, but when you do multiple tests, the **overall (family-wise) error rate increases**.

The combined probability of making at least one Type I error is:

$$1-(1-\alpha)^k$$

where:

- α=0.05
- k=6 test

Using that:

$$1-(1-0.05)^6 = 0.265$$

So the overall significance level becomes about **26.5%**, not 5%.

Uses of ANOVA

- **Comparing means of multiple groups**
  ANOVA is used to determine whether there are statistically significant differences between the means of three or more groups at the same time.
- **Testing treatment effects in experiments**
  It helps evaluate whether different treatments (e.g., medicines, teaching methods, fertilizers) produce different outcomes.
- **Analyzing variability in data**
  ANOVA separates total variation into **between-group** and **within-group** variation, helping researchers understand sources of differences.
- **Reducing Type I error**
  Instead of performing multiple t-tests, ANOVA controls the overall significance level, reducing the risk of false positive results.
- **Supporting decision-making in research and industry**
  It is widely used in fields like agriculture, business, psychology, and medicine to compare performance, quality, or effectiveness across groups.

We will study Comparing Mean for multiple group for this purpose ANOVA can be classified as:

One-Way ANOVA

One-way ANOVA is a statistical method used to compare the means of three or more groups based on **one factor**, for example, comparing a teacher's performance across different sections (A, B, C).

One-Way ANOVA Example (Teacher Performance by Sections)

Section	Student Scores	Mean Score
A	70, 75, 80	75
B	80, 85, 81	82
C	76, 78, 80	78

Two-Way ANOVA

Two-way ANOVA is used to compare group means based on **two factors** and to study their interaction, for example, comparing a teacher's performance across different sections (A, B, C) and teaching methods (lecture vs activity-based).

Two-Way ANOVA Example (Sections + Teaching Method)

Section	Lecture Method	Activity-Based Method
A	72, 75	78, 80
B	80, 82	85, 87
C	76, 78	79, 81

Data Table

Random samples of size $n_1, n_2 \ldots, n_t$ are drawn from the respective t populations. The data would have the following format:

Population	Data				Mean
1	y_{11}	y_{12}	...	y_{1n_1}	$\bar{y}_{1.}$
2	y_{21}	y_{22}	...	y_{2n_2}	$\bar{y}_{2.}$
⋮	⋮	⋮	⋮	⋮	⋮
t	y_{t1}	y_{t2}	...	y_{tn_t}	$\bar{y}_{t.}$

t: The total number of groups

y_{ij}: The j^{th} observation from the i^{th} population.

n_i: The sample size from the i^{th} population.

n_T: The total sample size: $n_T = \sum_{i=1}^{t} n_i$.

$\bar{y}_{i.}$: The mean of the sample from the i^{th} population.

$\bar{y}_{..}$: The mean of the combined data. Also called the overall mean.

Recall that we want to examine the between group variation and the within group variation. We can find an estimate of the variations with the following:

Sum of Squares for Treatment or the Between Group Sum of Squares

$$\text{SST} = \sum_{i=1}^{t} n_i(\bar{y}_{i.} - \bar{y}_{..})^2$$

Sum of Squares for Error or the Within Group Sum of Squares

$$\text{SSE} = \sum_{i,j}(y_{ij} - \bar{y}_{i.})^2$$

Total Sum of Squares

$$\text{TSS} = \sum_{i,j}(y_{ij} - \bar{y}_{..})^2$$

It can be derived that $\text{TSS} = \text{SST} + \text{SSE}$.

We can set up the ANOVA table to help us find the F-statistic. Hover over the light bulb to get more information on that item.

The ANOVA Table

Source	Df	SS	MS	F
Treatment	$t-1$	SST	$\text{MST} = \frac{\text{SST}}{t-1}$	$\frac{\text{MST}}{\text{MSE}}$
Error	$n_T - t$	SSE	$\text{MSE} = \frac{\text{SSE}}{n_T - t}$	
Total	$n_T - 1$	TSS		

The Critical value is found using the F-statistic and the F-distribution.. You will only need to know how to interpret it. If the F -Computed is less than our predetermined F tabular, we will reject the null hypothesis that all the means are equal.

The ANOVA table can easily be obtained by statistical software and hand computation of such quantities are very tedious.

Easy formulas:

1) Take sum of all values
2) Take sum of Square values
3) Correction Factor C.F=$\frac{(\sum X)^2}{n}$
4) Total variation :$\sum X^2$ – C.F
5) Between sample :$\frac{\sum X_{.j}^2}{r}$ – C.F
6) Compute F.

General procedure of hypothesis testing with ANOVA

H_o: $\mu_1=\mu_2=....=\mu_k$(all means are equal)

H_1: Not all k means are equal.(all means are different)

Level of Significance: α=

Test Statistics:

$$F = \frac{MS_b^2}{MS_w^2}$$

Critical region: F $\geq F_{\alpha(k-1,n-k)}$, k= no group, n= total observation in data

Computation:

1) Take sum of all values
2) Take sum of Square values
3) Correction Factor C.F=$\frac{(\sum X)^2}{n}$
4) Total variation :$\sum X^2$ – C.F
5) Between group :$\frac{\sum X_{.j}^2}{r}$ – C.F
6) Error/ within: Total variation – Between Sample Variation
7) Anova

Source variation	Df	Sum of Squares (SS)	Mean Square (MS)	F
Between Group	k-1	Step 5	SSb/df(b)	MSb/MSw
Within Group/ error	n-k	Step 6	SSW/df(w)	
Total	n-1	Step 4		

Conclusion:

Exercise 12.1 Elementary Statistics By Bluman

Question# **12. Weight Gain of Athletes** A researcher wishes to see whether there is any difference in the weight gains of athletes following one of three special diets. Athletes are randomly assigned to three groups and placed on the diet for 6 weeks. The weight gains (in pounds) are shown here. At α=0.05, can the researcher conclude that there is a difference in the diets?

Diet A	Diet B	Diet C
3	10	8
6	12	3
7	11	2
4	14	5
	8	
	6	

H_o: $\mu_1=\mu_2=\mu_3$(all three groups have same mean)

H_1: Not all k means are equal. (all three groups have different mean)

Level of Significance: α=0.05

Test Statistics:

$$F = \frac{MS_b^2}{MS_w^2}$$

Critical region: F $\geq F_{\alpha(k-1,n-k)}$ ➔ F $\geq F_{0.05(3-1,14-3)}$ ➔ F $\geq F_{0.05(2,11)}$ ➔ F ≥3.98

If CR satisfied reject Ho

Diet A	Diet B	Diet C
3	10	8
6	12	3
7	11	2
4	14	5
	8	
	6	

Computation:

1) Take sum of all values
2) Take sum of Square values
3) Correction Factor C.F=$\frac{(\sum X)^2}{n}$
4) Total variation :$\sum X^2$ – C.F
5) Between sample :$\frac{\sum X_{.j}^2}{r}$ – C.F
6) Error : Total variation – Between Sample Variation
7) Anova

Xij= ith row & jth column

X31= 3rd row 1st column

X41= 4th row 1st colum

X.1➔ 1st column sum

X1. ➔ 1st row sum

	Diet A		Diet B		Diet C	
	X	X^2	X	X^2	X	X^2
	3	9	10	100	8	64
	6	36	12	144	3	9
	7	49	11	121	5	25
	4	16	14	196	2	4
			8	64		
			6	36		
Sum	20	110	61	661	18	102

3) Correction Factor C.F=$\frac{(\sum X)^2}{n}$= $\frac{(20+61+18)^2}{14} = 700.0714$
4) Total variation :$\sum X^2$ – C.F= (110+661+102)-700.0714=172.9286
5) Between group :$\frac{\sum X_{.j}^2}{r}$ – C.F = $\left(\frac{20^2}{4} + \frac{61^2}{6} + \frac{18^2}{4}\right) - 700.0714$ =101.0953
6) Within group: total variation – between group = 172.9286-101.0953=71.8333

Source variation	Df	Sum of Squares (SS)	Mean Square (MS)	F
Between Group	k-1	Step 5	SSb/df(b)	MSb/MSw
Within Group/ error	n-k	Step 6	SSW/df(w)	
Total	n-1	Step 4		

Source of variation	d.f	Sum Of Square	Mean Square	Computed f
Between group	3-1=2	101.0953	(101.0953/2)=50.5476	50.5476/6.5303= 7.740
Within group	14-3=11	71.8333	(71.8333/11)=6.5303	
Total variation	14-1=13	172.9286		

7.740>3.98 ➔ satisfied

Conclusion: As Cr satisfied therefore there is evidence to reject Ho at 0.05 level of significance we conclude that all three diet have different weight gains.

Example:

Given the data below, test the hypothesis that the means life of three medicine are equal. Let α =0.05.

Medicine 1	Medicine 2	Medicine 3
40	70	45
50	65	38
60	66	60
65	50	42

Solution:

H_o: $\mu_1=\mu_2=\mu_3$

H_1: all 3 means are not equal.

Level of Significance: α=0.05

Test Statistics:

$$F = \frac{MS_b^2}{MS_w^2}$$

Critical region: F $\geq F_{\alpha(k-1,n-k)}$ ➔ F $\geq F_{0.05(3-1,12-3)}$ ➔ F $\geq F_{0.05(2,9)}$ ➔ F ≥4.26

If CR satisfied reject Ho

Computation:

	Medicine 1		Medicine 2		Medicine 3	
	X	X^2	X	X^2	X	X^2
	40		70		45	
	50		65		38	
	60		66		60	
	65		50		42	
Total	215	11925	251	15981	185	8833

3) Correction Factor C.F=$\frac{(\sum X)^2}{n}$=$\frac{(215+251+185)^2}{12} = 35316.75$
4) Total variation :$\sum X^2$ – C.F=(11925+15981+8833)-35316.75=1422.25
5) Between sample :$\frac{\sum X_j^2}{r}$ – C.F = $\left(\frac{215^2}{4} + \frac{251^2}{4} + \frac{185^2}{4}\right) - 35316.75$= 546
6) With sample: total variation – between sample = 1422.25-546=876.25

Source of variation	d.f	Sum Of Square	Mean Square	Computed f
Between samples	2	546	546/2=273	273/97.3611=2.80
Within Samples	9	876.25	876.25/9=97.3611	
Total variation	11	1422.25		

2.80>4.26➔ doesn't satisfied

Conclusion: As Cr doesn't satisfied therefore there is no evidence to reject Ho at 0.05 level of significance we conclude that all 3 medicine have equal life time.

2) Three sections of the same elementary mathematics course are taught by 3 teachers. The final grades were recorded as below table. Test the hypothesis that average grades given by the three teachers are equal. Use a 0.05 level of significance.

A	75	91	84	45	82	75	68	47	95	38
B	59	83	99	77	65	81	34	81	77	88
B	94	51	82							
C	66	77	51	90	73	90	71	68	69	

Solution:

H_o: $\mu_1=\mu_2=\mu_3$

H_1: Not all 3 means are equal.

Level of Significance: α=0.05, degree of freedom= (k-1), (n-k) = 2,29

Test Statistics:

$$F = \frac{S_b^2}{S_w^2}$$

Critical region: F $\geq F_{\alpha(k-1,n-k)}$

Computation:

Obs	A	A2	B	B2	C	C2
1	75		59		66	
2	91		83		77	
3	84		99		51	
4	45		77		90	
5	82		65		73	
6	75		81		90	
7	68		34		71	
8	47		81		68	
9	95		77		69	
10	38		88			
11			94			
12			51			
13			82			
Total						

1) Correction Factor C.F= $\frac{(\sum X)^2}{n}$=
2) Total variation :$\sum X^2$ – C.F=
3) Between sample :$\frac{\sum X_{.j}^2}{r}$ – C.F =
4) Within sample: total variation – between sample =

Source of variation	d.f	Sum Of Square	Mean Square	Computed f

Between samples				
Within Samples				
Total variation				

Conclusion: